Yash Raj Shrivastava

Co-founder & Full-Stack Developer | B.Tech CSE (IILM, 2025–2029)
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Summary

- Co-founder, product designer, and full-stack developer. Shipped web applications in Python and JavaScript, robust data models with Prisma, and took a reservation/analytics SaaS from prototype to deployment.
- Hardware portfolio: open-source ESP32 quadruped robot and ROS2 swarm-drone system — embedded systems plus AI/ML.
- Completed the 360 Labs AI test work week; Sagar Defence Engineering technical interview (drone/AI internship).

Experience

Co-founder & Product Designer — Stealth-mode SaaS **(ongoing)**

- Leading product strategy, technical architecture, and investor outreach for a reservation and analytics SaaS platform.
- Owns the full product lifecycle: ideation, UX design, full-stack development, and deployment.

Lead Developer — Gurtai Bot **(ongoing)**

- Built a Python-based Telegram AI assistant with local LLM integration via Ollama for privacy-first, zero API-cost responses.
- Designed a plugin-based modular architecture for extensible workflow automation; containerized with Docker.

Software Projects

CamPass — Student Safety & Tracking **(Flutter, Node.js, PostgreSQL, FCM)**

- Real-time geofence tracking, SOS alerts, and smart notifications via Firebase Cloud Messaging; digital pass system with barcode scanning and role-based access.

RiceScan — AI Crop-Disease Detection **(Python, EfficientNetV2S)**

- AI-powered agricultural app (EfficientNetV2S) with local knowledge base for symptoms/treatments and a feedback loop for model retraining.

Hardware & Robotics Projects

Sesame Robot — ESP32 8-DOF Quadruped (Open Source)

- Built a mini quadruped from scratch: ESP32 DevKitC, 8× MG90S metal-gear servos, 3D-printed PLA chassis; custom firmware (Wi-Fi AP, captive-portal browser control, REST API, OLED status).
- Pinned ESP32Servo v3.0.9, configured PWM timer allocation for 8 servos; 4 A power failure root-caused and fixed via firmware ECO mode + upgraded 5 A buck (verified 2 A).
- Basic walk gait achieved; Gazebo/ROS2 simulation in WSL2; open source at github.com/Yashop965/sesame-robot.

Swarm Drone System — ROS2 Humble Simulation

- Carrier hexacopter carrying 4–6 autonomous sub-drones; capability-aware task scoring, ROS→MQTT/TCP mesh bridge, Gazebo physics with 4/4 validated task assignments (clean colcon build, 1.29 s).
- Full stack: ground-station UI → mother drone → sub-drone mesh → return-and-dock cycle with failsafe navigation.

Other Hardware — DIY Cameras & Drone Component Testing

- Custom camera prototypes with controlled lighting and exposure-control systems; drone ESC (30 A, Hobbywing, DShot) and motor-driver testing for flight stability and power efficiency.

Experiments & Ops

- **Simulation:** ROS2 Humble + Gazebo headless testing in WSL2 — URDF modeling, gait-pattern validation (4/4 task assignments on the swarm system).
- **Ops:** Docker for reproducible environments (Gurtai, dev fleets); CI/CD failure triage (local-pass/CI-fail); E2E test automation (Playwright).
- **Lab work:** IoT & Robotics Foundation Lab (CSE25309) — ESP32/I2C/PWM servo control, circuit simulation (Tinkercad), 8-DOF quadruped project.

Skills

- **Languages:** Python, JavaScript, TypeScript, C++, Dart, SQL, Bash
- **AI & ML:** GenAI applications, prompt engineering, LLMOps, AI agents & chatbots, OpenCV, EfficientNet (transfer learning), RAG concepts

- **Web & Mobile:** Next.js, React, Node.js, Express, Flutter, Prisma/PostgreSQL, Firebase, Docker
- **Embedded & Hardware:** ESP32, Raspberry Pi, Arduino, ROS2 (Humble), Gazebo, servo/PWM/I2C, breadboard & schematic-level circuit work
- **Tools:** Git, GitHub Actions-style CI debugging, Vercel, ESLint, CAD (Fusion 360), WSL2, AWS (basics)

Education

- **IILM University, Gurugram** — B.Tech, Computer Science & Engineering (2025–2029)

Certifications

- DeepLearning.AI — *AI Python for Beginners*; *Build Rich-Context AI Apps with Anthropic*
- 360 Labs — AI Test Work Week (Aug 2026)